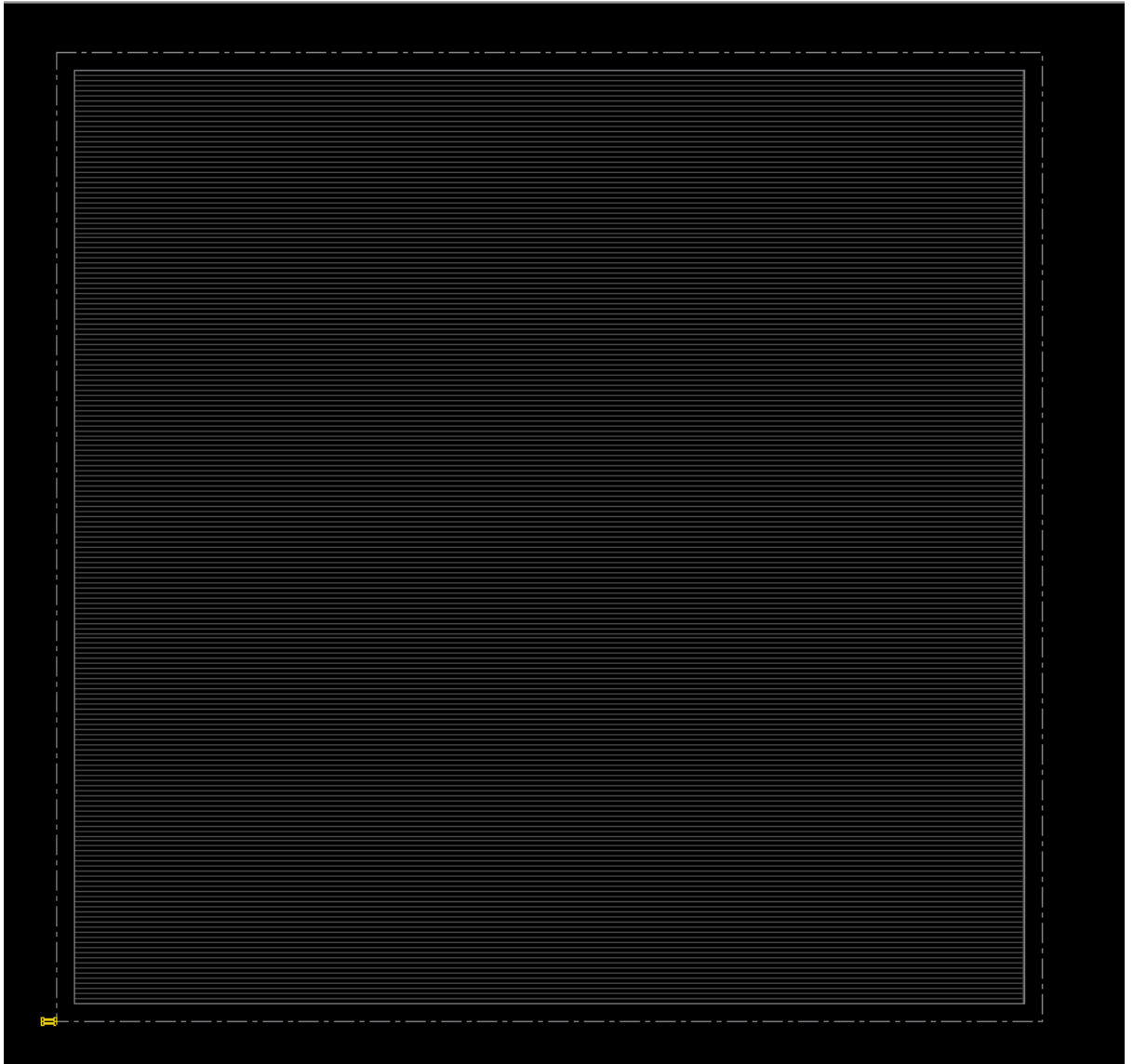
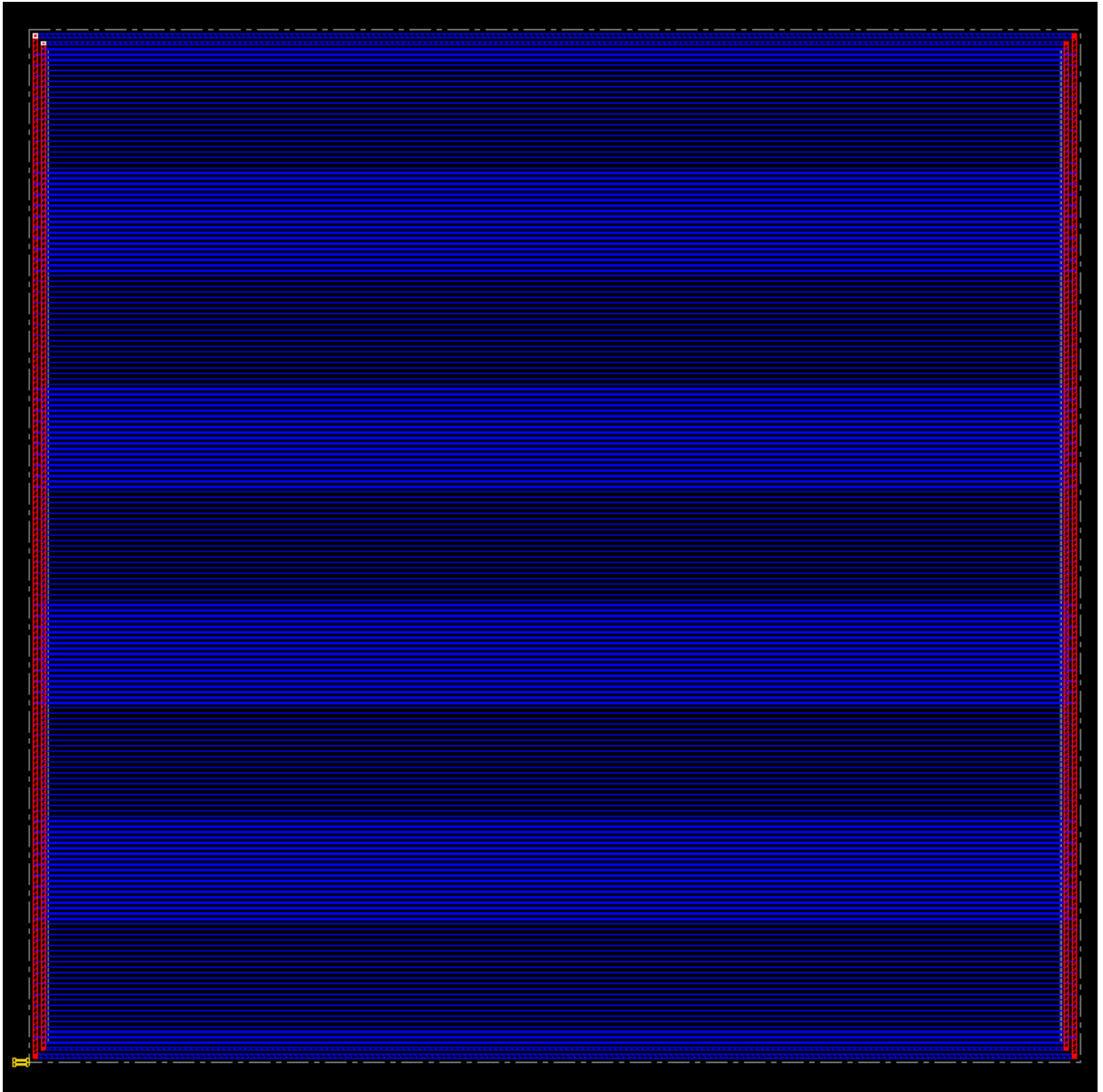


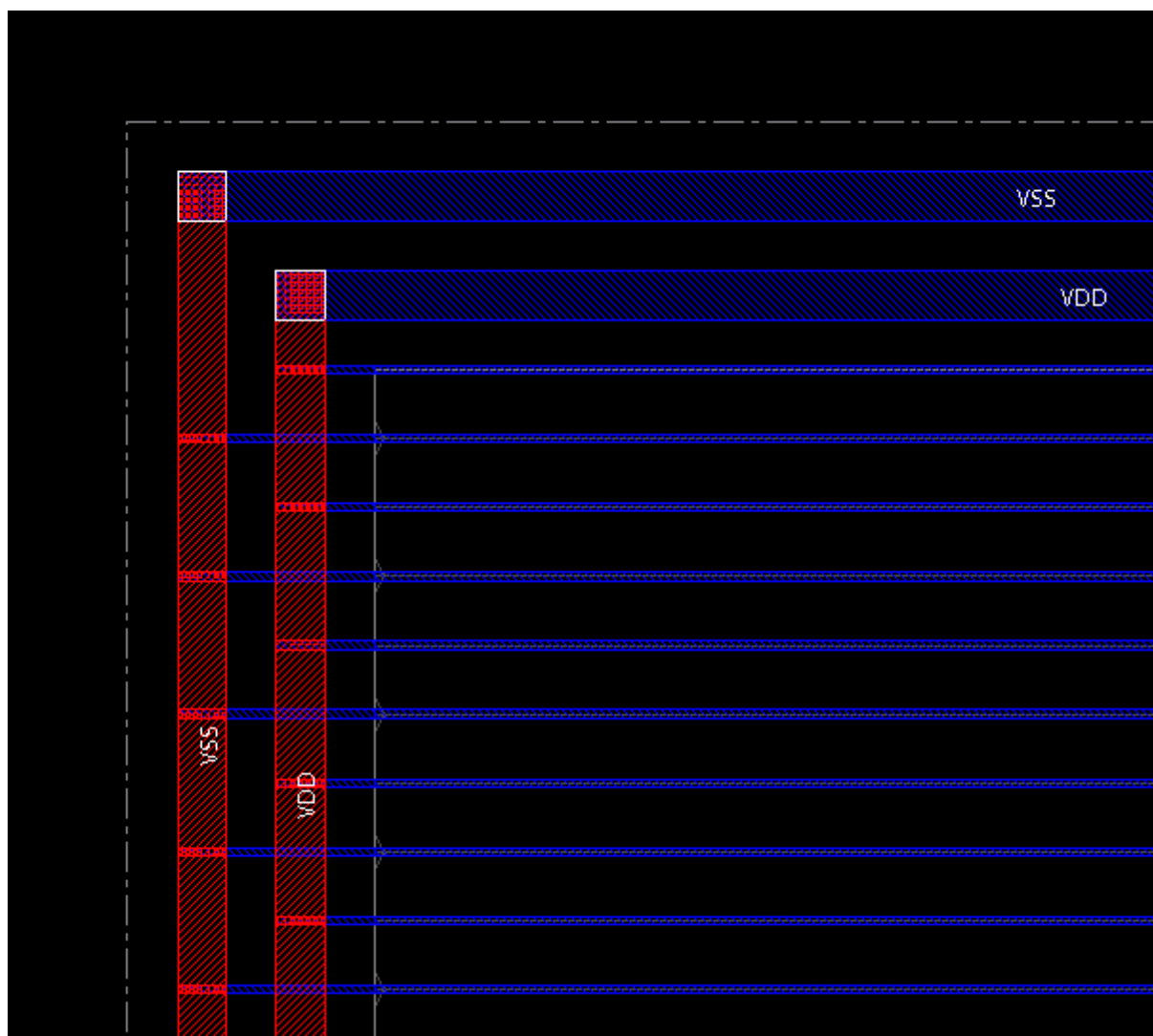
32-bit Pipelined Multiplier

1. chip outlining

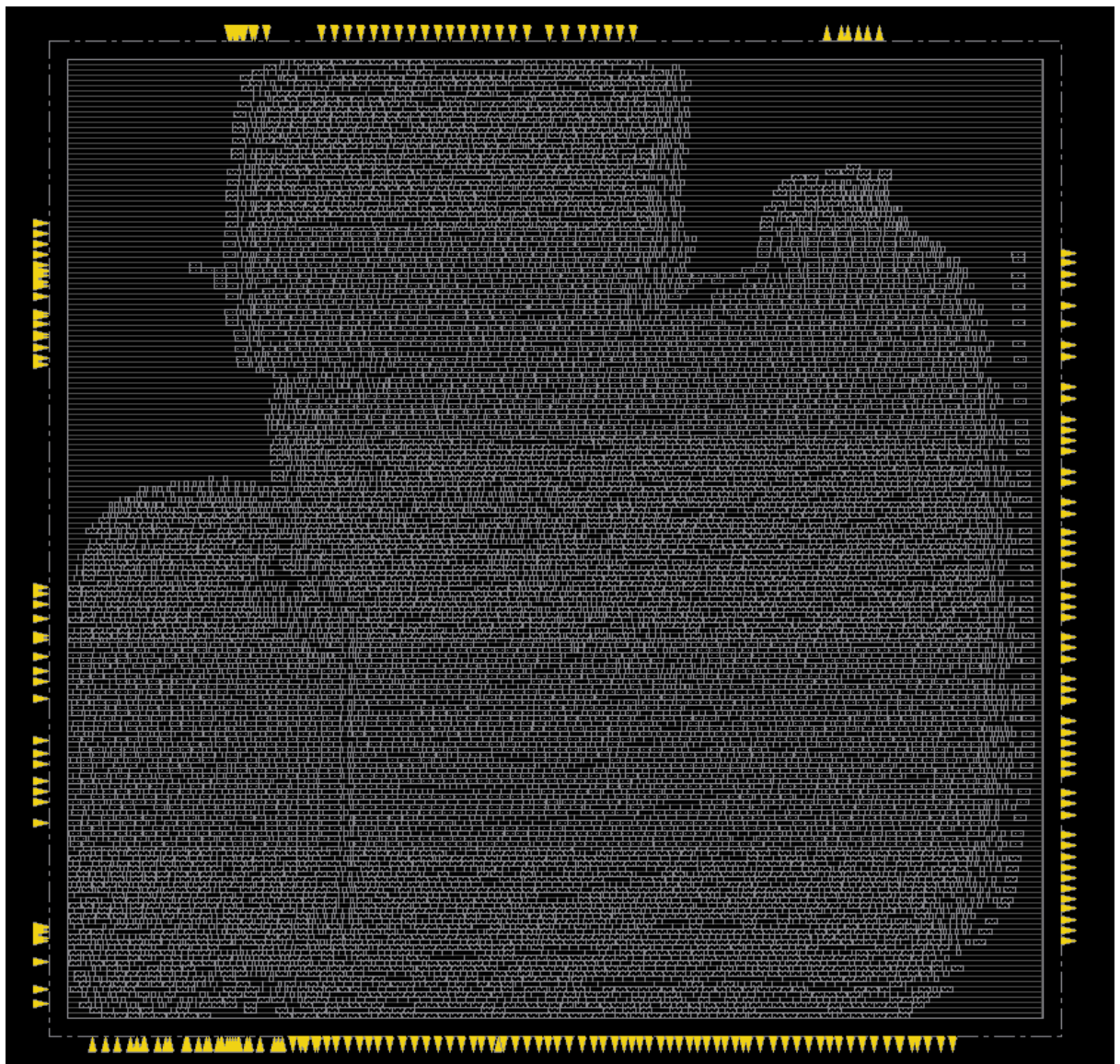


2. P/G network Design





3. Placement

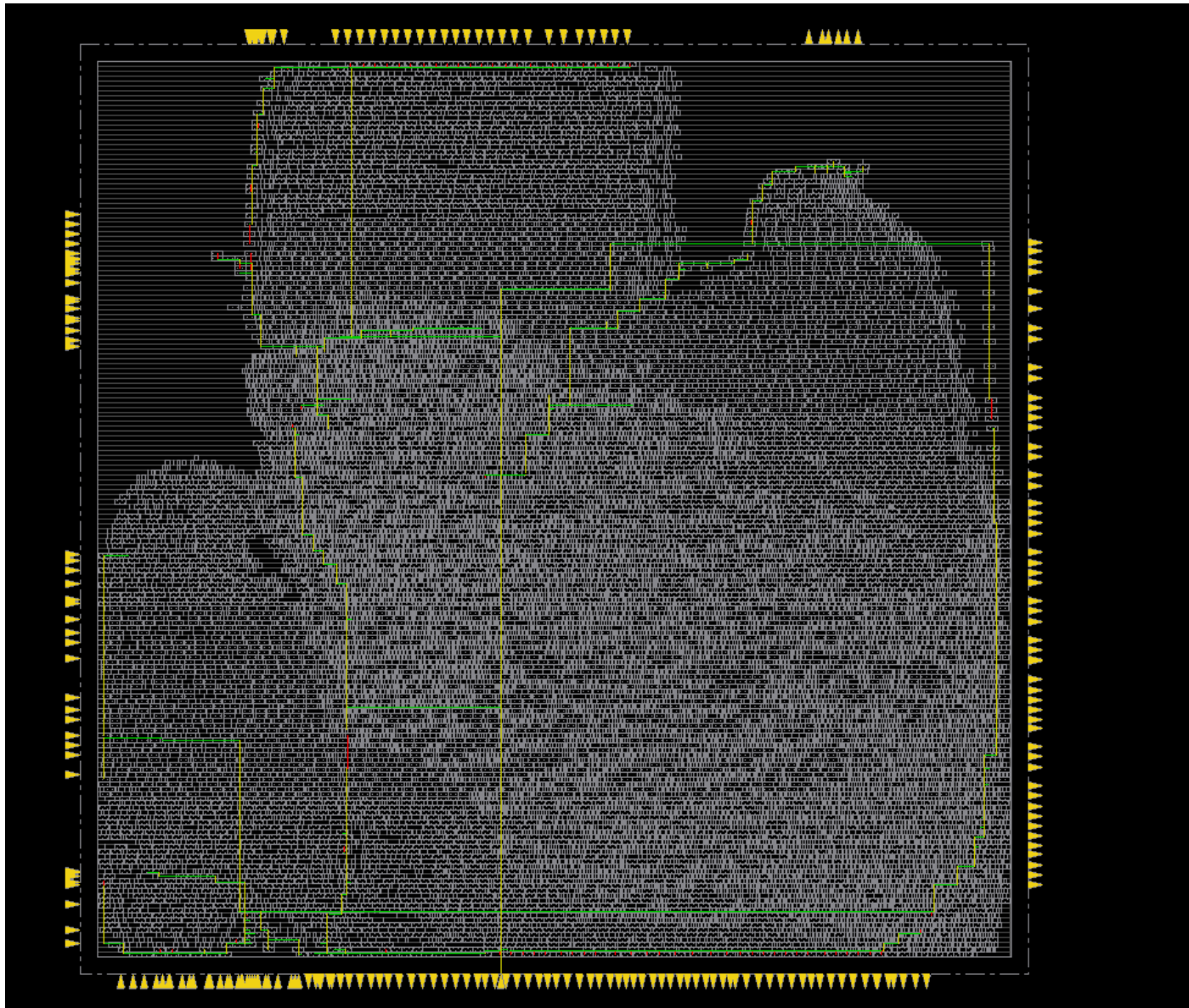


- a. WNS: -2.209 ns
- b. TNS: -97.350 ns
- c. Density: 48.563%
- d. Power: 25.416 uW
- e. Wire length: 133553 um

4. Pre-CTS Optimization

- a. WNS: 0.009 ns
- b. TNS: 0.0 ns
- c. Density: 64.399%
- d. Power: 28.608 μ W

5. CTS



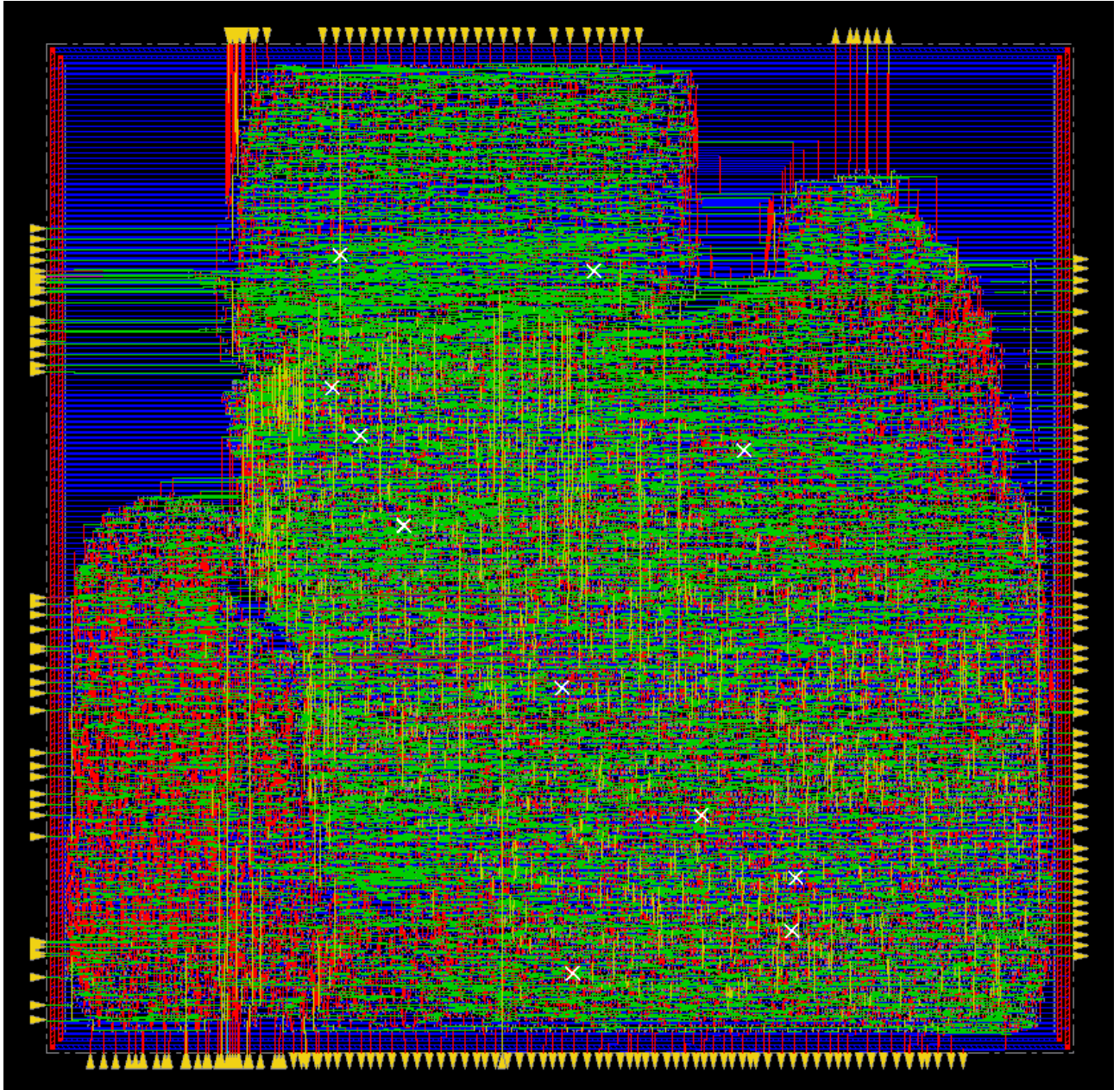
- a. WNS: 0.001 ns
- b. TNS: 0.0 ns
- c. Density: 66.919 %
- d. Power: 30.269 μ W

6. Post-CTS Optimization

- a. WNS: 0.0 ns

- b. TNS: 0.0 ns
- c. Density: 66.890%
- d. Power: 30.263 μ W

7. Routing



- a. WNS: 0.016 ns
- b. TNS: 0.00 ns
- c. Density: 66.890%
- d. Power: 30.129 μ W
- e. Wire length: 170624 μ m

8. Post-route Optimization

- a. WNS: 0.015 ns

- b. TNS: 0.00 ns
- c. Density: 66.890%
- d. Power: 30.130 uW
- e. Wire length: 170986 um

9. Verification

a. verify DRC: 0 viols

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VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 55.080 54.000} 1 of 25
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {55.080 0.000 110.160 54.000} 2 of 25
VERIFY DRC ..... Sub-Area : 2 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {110.160 0.000 165.240 54.000} 3 of 25
VERIFY DRC ..... Sub-Area : 3 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {165.240 0.000 220.320 54.000} 4 of 25
VERIFY DRC ..... Sub-Area : 4 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {220.320 0.000 272.160 54.000} 5 of 25
VERIFY DRC ..... Sub-Area : 5 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 54.000 55.080 108.000} 6 of 25
VERIFY DRC ..... Sub-Area : 6 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {55.080 54.000 110.160 108.000} 7 of 25
VERIFY DRC ..... Sub-Area : 7 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {110.160 54.000 165.240 108.000} 8 of 25
VERIFY DRC ..... Sub-Area : 8 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {165.240 54.000 220.320 108.000} 9 of 25
VERIFY DRC ..... Sub-Area : 9 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {220.320 54.000 272.160 108.000} 10 of 25
VERIFY DRC ..... Sub-Area : 10 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 108.000 55.080 162.000} 11 of 25
VERIFY DRC ..... Sub-Area : 11 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {55.080 108.000 110.160 162.000} 12 of 25
VERIFY DRC ..... Sub-Area : 12 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {110.160 108.000 165.240 162.000} 13 of 25
VERIFY DRC ..... Sub-Area : 13 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {165.240 108.000 220.320 162.000} 14 of 25
VERIFY DRC ..... Sub-Area : 14 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {220.320 108.000 272.160 162.000} 15 of 25
VERIFY DRC ..... Sub-Area : 15 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 162.000 55.080 216.000} 16 of 25
VERIFY DRC ..... Sub-Area : 16 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {55.080 162.000 110.160 216.000} 17 of 25
VERIFY DRC ..... Sub-Area : 17 complete 0 Viols.

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VERIFY DRC Sub-Area: {110.160 162.000 165.240 216.000} 18 of 25
VERIFY DRC Sub-Area : 18 complete 0 Viols.
VERIFY DRC Sub-Area: {165.240 162.000 220.320 216.000} 19 of 25
VERIFY DRC Sub-Area : 19 complete 0 Viols.
VERIFY DRC Sub-Area: {220.320 162.000 272.160 216.000} 20 of 25
VERIFY DRC Sub-Area : 20 complete 0 Viols.
VERIFY DRC Sub-Area: {0.000 216.000 55.080 267.680} 21 of 25
VERIFY DRC Sub-Area : 21 complete 0 Viols.
VERIFY DRC Sub-Area: {55.080 216.000 110.160 267.680} 22 of 25
VERIFY DRC Sub-Area : 22 complete 0 Viols.
VERIFY DRC Sub-Area: {110.160 216.000 165.240 267.680} 23 of 25
VERIFY DRC Sub-Area : 23 complete 0 Viols.
VERIFY DRC Sub-Area: {165.240 216.000 220.320 267.680} 24 of 25
VERIFY DRC Sub-Area : 24 complete 0 Viols.
VERIFY DRC Sub-Area: {220.320 216.000 272.160 267.680} 25 of 25
VERIFY DRC Sub-Area : 25 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:02.0 ELAPSED TIME: 2.00 MEM: 0.0M) ***

b. verify connectivity: 0 viols

***** End: VERIFY CONNECTIVITY *****

Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.7 MEM: 0.000M)

	WNS(ns)	TNS(ns)	Density(%)	Power(mW)	WL(um)
Placement	-2.209	-97.350	48.563	25.416	133553
Pre-CTS opt	0.009	0.0	64.399	28.608	
CTS	0.001	0.0	66.919	30.269	
Post-CTS opt	0.0	0.0	66.890	30.263	
Routing	0.016	0.0	66.89	30.129	170624
Post-Route opt	0.015	0.0	66.890	30.130	170986